

Deep Learning for a Regression Problem

1 Introduction

You are tasked with using deep learning to solve a housing price prediction problem. The dataset to be processed is available on Moodle, it is called "housing_prices_data.csv".

2 Steps

2.1 Step 1: Importing the CSV File

```
import pandas as pd
Import the "housing_prices_data.csv" CSV file
Read the csv file
```

2.2 Step 2: Exploring the Dataset

Use the pandas package to read and describe the dataset.

2.3 Step 3: Converting the Dataset to NumPy Arrays

Convert the dataset into NumPy arrays for processing.

2.4 Step 4: Splitting the Dataset

The first 10 columns correspond to the features. Split the dataset into input data (input features) and labels.

2.5 Step 5: Scaling the Data

Scale the data so that they have similar orders of magnitude using a function from scikit-learn.

2.6 Step 6: Partitioning the Dataset

Partition the dataset into a training set, a validation set, and a test set using the `train_test_split` function from scikitlearn.

2.7 Step 7: Displaying the Data

Display the resulting data.

2.8 Step 8: Building the Deep Network

Build a deep network with 2 hidden layers of 32 neurons each, with ReLU activation functions. Explain the complete architecture up to the output layer.

2.9 Step 9: Compiling the Model

Compile the model with an optimizer of your choice and an appropriate loss function. Explain your choices.

2.10 Step 10: Training the Model

Train the model by varying the number of epochs.

2.11 Step 11: Evaluating the Model

Evaluate your model.

2.12 Step 12: Visualizing Losses

Visualize the losses for the training data and the validation data using a graph with matplotlib.